CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 2: Algorithms, Programming and Logic | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 07: ALGORITHM DESIGN AND PROBLEM SOLVING

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/22/M/J/24 Q1 & Q3 (2024 Series)] [Total: 6 marks]

(a) List the four main stages of the Program Development Life Cycle (PDLC) in correct order. **[2 marks]**

(b) Differentiate between verification and validation of data input. **[2 marks]**

(c) Give two validation checks and describe what each check tests for. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) Analysis -> Design -> Coding -> Testing [2 marks].
(b) Validation: Software check to ensure data is sensible/reasonable [1 mark]. Verification: Check to ensure data entered matches original source without transcription errors [1 mark].
(c) Range check (within limits), Length check (correct number of characters), Format check (correct pattern), Presence check (not empty) [2 marks].

Question 2 [2210/22/M/J/24 Q4 (2024 Series)] [Total: 6 marks]

(a) An algorithm is represented by the following pseudocode:

```
Total <- 0
FOR Count <- 1 TO 5
   INPUT Num
   IF Num > 10 THEN
      Total <- Total + Num
   ENDIF
NEXT Count
OUTPUT Total
```

Complete the trace table for the inputs: 12, 5, 15, 8, 20. **[4 marks]**

Count	Num	Num > 10	Total	OUTPUT
			0	
1	12	TRUE	12	
2	5	FALSE	12	

3	15	TRUE	27	
4	8	FALSE	27	
5	20	TRUE	47	
				47

(b) State three types of test data used when testing an algorithm and provide an example of each for an age input between 11 and 18. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) Trace table completed accurately with final Total = 47 and OUTPUT 47 [4 marks].
(b) Normal data: 14; Boundary data: 11, 18; Erroneous data: 25, "ten" [2 marks].

Question 3 [2210/22/O/N/23 Q3 (2023 Series)] [Total: 6 marks]

(a) Describe how the Bubble Sort algorithm sorts an array of numbers into ascending order.**[3 marks]**

(b) Compare Linear Search and Binary Search in terms of prerequisites and efficiency. **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) Bubble sort compares adjacent pairs of elements; if out of order, swaps them [1 mark].
Iterates through array repeatedly until a complete pass occurs with zero swaps [2 marks].
(b) Binary search requires the array to be pre-sorted, while linear search works on unsorted arrays [1.5 marks].
Binary search is significantly faster / more efficient for large lists because it halves the search space each step [1.5 marks].
